CHETAN SRIKANT MANDIGA

SOFTWARE ENGINEER | APPLIED AI, COMPUTER VISION & FULL-STACK SYSTEMS

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PROFILE

Computer Science graduate and systems engineer with experience across enterprise identity operations, AI-backed data applications, computer vision, and full-stack product development. Builds practical Python workflows using OpenCV, NumPy, PyTorch, TensorFlow/Keras, and scikit-learn, and turns model outputs into usable Streamlit or web interfaces. Recognized as part of the winning AI4TG public-transport solution team.

CORE COMPETENCIES

Programming: Python, Java, C++, JavaScript; Data Structures & Algorithms, OOP, debugging, system-design fundamentals
AI & data: NumPy, Pandas, scikit-learn, PyTorch, TensorFlow/Keras, forecasting, classification, feature engineering, model evaluation
Computer vision: OpenCV, YOLOv11, Detectron2, Faster R-CNN, U-Net, CNNs; enhancement, filtering, edges, contours, segmentation, feature extraction
Applications & platforms: React, Node.js/Express, Streamlit, Firebase, REST APIs, MongoDB, MySQL, Git, Active Directory, Azure AD

PROFESSIONAL EXPERIENCE

Systems Engineer — IAM Cloud (Active Directory / Azure AD) **Aug 2025 – Present**
HCLTech *Noida, India*

- Manage enterprise Active Directory and Azure AD workflows spanning user provisioning, access control, and hybrid-identity integration.
- Troubleshoot authentication failures, account lockouts, directory synchronization, and access issues in production using structured root-cause analysis.
- Support reliable identity operations through incident ownership, documentation, and coordination across operational teams.

Full Stack Data Scientist **Mar 2025 – Aug 2025**
Innomatics Research Labs *Hyderabad, India*

- Built AI-backed applications for enterprise forecasting and TGSRTC public-transport demand analysis, connecting model outputs to interactive dashboards and stakeholder-facing workflows.
- Contributed substantially to the web application and visual-analysis layer of the winning Telangana AI Rising / AI4TG transport solution; made forecast outputs easier to inspect and communicate.
- Worked with forecasting models, a custom chatbot, CRISP-ML(Q), REST-style integration, and Git-based delivery practices across two end-to-end data platforms.

Project Intern **May 2023 – Jul 2023**
DRDO — Defence Research and Development Organisation *Hyderabad, India*

- Designed Python/Tkinter interfaces for RS-422 device communication, improving operator workflows and reducing communication errors during testing.

EDUCATION & RECOGNITION

B.Tech — Computer Science and Engineering **2021 – 2025**
BML Munjal University *India*
Minor in Entrepreneurship, 2023–2025 | Academic work in Digital Image Processing and Machine Learning
Winner, AI4TG Grand Challenge / Telangana AI Rising — public-transport utilization solution (team recognition)

SELECTED APPLIED AI & COMPUTER VISION PROJECTS

TGSRTC Public Transport Demand Platform | *Forecasting, dashboards, decision support* **2025**

- Contributed to the web application that surfaced public-transport demand forecasts through visual exploration and stakeholder-oriented interpretation; the team won the AI4TG Grand Challenge.
- Focused on making forecast outputs understandable and explorable in a collaborative challenge setting; does not represent a solo-built or operationally deployed system.

Computer Vision Image Search | *Python, YOLOv11, Streamlit* **2025**

- Built an end-to-end visual-retrieval workflow that detects objects in image collections, converts detections into a searchable label index, matches text queries, ranks results, and presents retrieved images in an interactive interface.
- Designed the pipeline as separate detection, indexing, query-matching, ranking, and presentation stages, making detector coverage and ambiguous-label limitations explicit.

Conversational RAG Application | *Streamlit, LangGraph, Hugging Face, Groq* **2025**

- Developed a multi-turn, document-grounded assistant with explicit retrieval and generation stages, embedding-based search, maintained conversation state, and an interface for inspecting grounded responses.
- Used LangGraph state to keep retrieval and response generation modular while Groq inference supplied the language-model layer.

Driver Drowsiness Detection | *OpenCV, NumPy, TensorFlow/Keras* **2024**

- Created a real-time camera prototype that detects face and eye regions, interprets visual signals, and triggers alerts; optimized the frame-processing path while documenting that the prototype is not a certified safety system.
- Combined traditional frame processing with learned inference to demonstrate an end-to-end camera-to-alert workflow.

Detection & Segmentation Experiments | *PyTorch, YOLOv11, Detectron2, Faster R-CNN, U-Net* **2024–2025**

- Implemented notebook-based experiments in custom object detection, instance segmentation, Faster R-CNN, Detectron2, and U-Net segmentation, with emphasis on dataset preparation, inference inspection, and model limitations.

Digital Image Processing Laboratory | *OpenCV, NumPy, Python* **2023–2025**

- Implemented enhancement, CLAHE and histogram equalization, denoising and filtering, color thresholding, affine and perspective transforms, contours, edge detection, morphology, region processing, resizing, and segmentation in reproducible notebooks.

SELECTED PRODUCT & FULL-STACK PROJECTS

ZAP Ride-Hailing Platform | *React, Express, MongoDB, JWT* **2025**

- Built an end-to-end ride-booking product covering account flows, vehicle selection, trip requests, protected routes, and persistent backend data through a React client and Express API.

LOOP Real-Time Chat | *React, Firebase, React Router* **2024**

- Developed an authenticated messaging application with shared state, real-time conversations, route-based navigation, and media support using Firebase services.

Identifying Creditworthy Borrowers | *scikit-learn, Streamlit, NumPy* **2023**

- Created a classification workflow that prepares borrower data, generates creditworthiness predictions, and exposes results through a review-oriented Streamlit interface.

ADDITIONAL TECHNICAL BREADTH

Additional academic and exploratory work includes radar-threat classification, medical-image classification, eye segmentation, stereo-depth estimation, sentiment analysis, and voice-driven ordering prototypes. These are presented as notebook experiments rather than production deployments.

Public code portfolio: github.com/ChetanSrikant